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Кафедра «Архитектура и цифровые технологии»

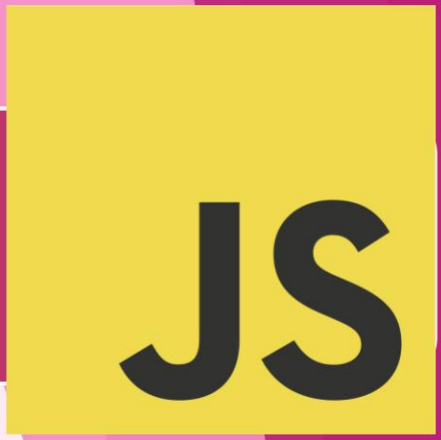
Тема 8 Выражения и операции JavaScript

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Ташкент 2025

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План:



8.1 Работа с объектами

8.2 Операции JavaScript

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💻 JavaScript: Expressions and Operations

🔷 8.1 Working with Objects

🧩 What is an Object?

An **object** in JavaScript is a **collection of data** or **information** stored as **key–value pairs**.

It helps to **group related information** together.

For example, a person, a car, or a student can be an object.

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🧠 Example:

javascript

Copy code

```javascript
let person = {
  name: "John",
  age: 25,
  country: "USA"
};
```

Here:

- `name`, `age`, and `country` are **keys (properties)**.
- `"John"`, `25`, and `"USA"` are **values**.

So this object means:

> John is 25 years old and lives in the USA.

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🔷 Accessing Object Values

We can get data from an object in two ways:

1. Dot notation

javascript Copy code

```
console.log(person.name);
```

👉 Output: `John`

2. Bracket notation

javascript

```
console.log(person["age"]);
```

👉 Output: `25`



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◆ Adding New Properties

We can add new data to an object:

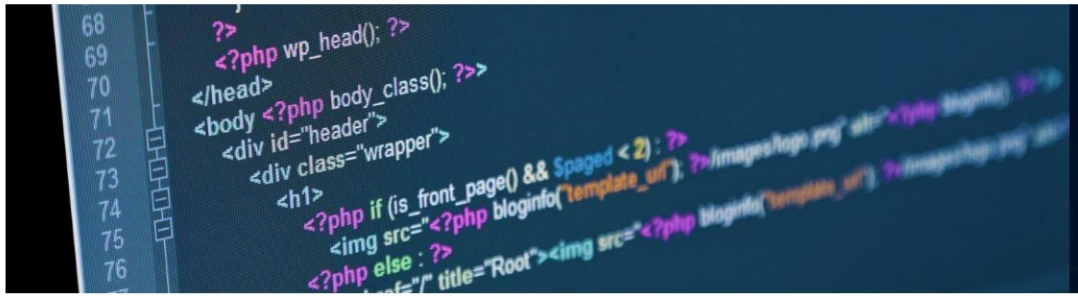
javascript Copy code

```
person.job = "Teacher";
```

Now the object becomes:

javascript Copy code

```
{ name: "John", age: 25, country: "USA", job: "Teacher" }
```





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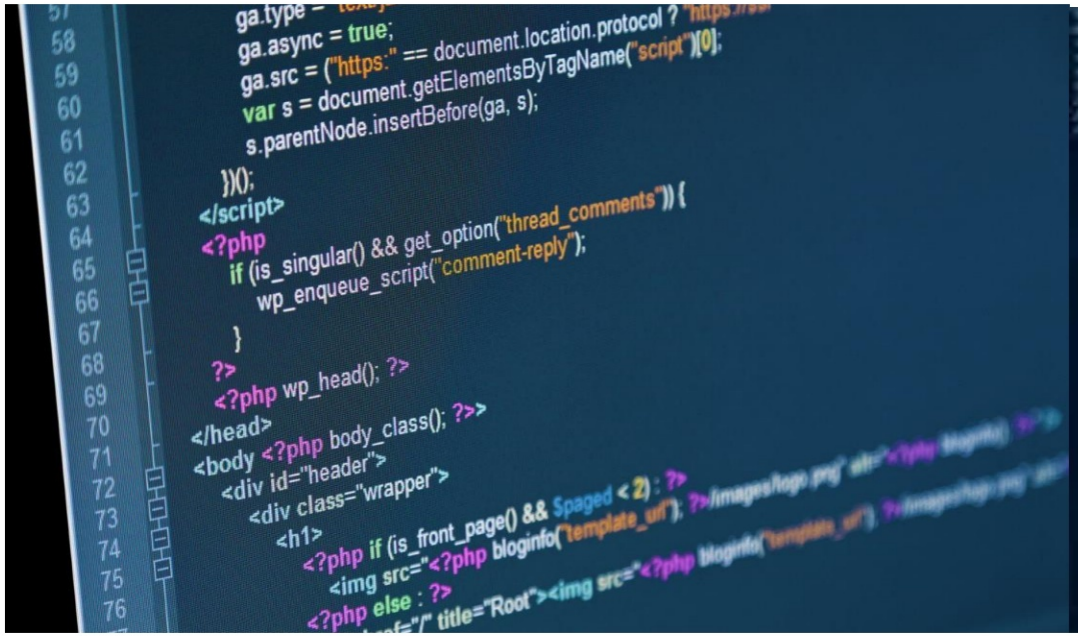
◆ Changing Property Values

javascript

Copy code

```
person.age = 26;
```

Now John’s age is 26.





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◆ Deleting Properties

javascript

Copy code

```
delete person.country;
```

Now `country` is removed from the object.



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🔷 Example: Object with a Function (Method)

Objects can also have **functions (methods)**.

javascript

Copy code

```javascript
let car = {
  brand: "Toyota",
  color: "black",
  start: function() {
    console.log("The car is starting...");
  }
};

car.start(); // Output: The car is starting...
```

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🔷 8.2 JavaScript Operations (Operators)

Operations in JavaScript are actions performed on values and variables.

They are written using **operators** like `+`, `-`, `*`, `/`, `=`, `==`, etc.

🧮 1. Arithmetic Operators

Used for **mathematical calculations.**

Operator	Meaning	Example	Result
`+`	Addition	`5 + 3`	8
`-`	Subtraction	`10 - 4`	6
`*`	Multiplication	`6 * 2`	12
`/`	Division	`8 / 2`	4
`%`	Modulus (remainder)	`10 % 3`	1
`**`	Power	`2 ** 3`	8

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💬 2. String Operator

When you use `+` with **strings**, it joins them together.

This is called **concatenation**.

javascript

Copy code

```javascript
let firstName = "Ali";
let lastName = "Khan";
console.log(firstName + " " + lastName);
```

👉 Output: `Ali Khan`

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3. Assignment Operators

Used to assign values to variables.

Operator	Example	Same as
`=`	`x = 5`	x = 5
`+=`	`x += 3`	x = x + 3
`-=`	`x -= 2`	x = x - 2
`*=`	`x *= 4`	x = x * 4
`/=`	`x /= 2`	x = x / 2

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⚖️ 4. Comparison Operators

Used to compare two values.

They return **true** or **false**.

Operator	Meaning	Example	Result
`==`	equal to	`5 == "5"`	true
`===`	equal value and type	`5 === "5"`	false
`!=`	not equal	`5 != 3`	true
`>`	greater than	`10 > 5`	true
`<`	less than	`3 < 8`	true
`>=`	greater or equal	`7 >= 7`	true
`<=`	less or equal	`4 <= 6`	true

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🔀 5. Logical Operators

Used to combine conditions.

Operator	Meaning	Example	Result
`&&`	AND	`(5 > 3 && 6 > 2)`	true
`	`	`	OR
`!`	NOT	`!(5 > 3)`	false

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🧩 6. Increment and Decrement Operators

Used to add or subtract 1 from a variable.

javascript

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```javascript
let x = 5;
x++; // same as x = x + 1 → 6
x--; // same as x = x - 1 → 5
```

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💡 Example combining everything:

javascript

Copy code

```javascript
let student = {
  name: "Sami",
  score: 80
};

student.score += 10;  // student.score = 90
if (student.score >= 90) {
  console.log(student.name + " passed the exam!");
} else {
  console.log(student.name + " failed.");
}
```

👉 Output: `Sami passed the exam!`

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Summary

Topic	Description
Object	A collection of key–value pairs (like a mini-database).
Access Object Data	Use `.` or `[]` notation.
Operators	Symbols to do math, compare, or assign values.
Main Operator Types	Arithmetic, Comparison, Logical, String, Assignment.